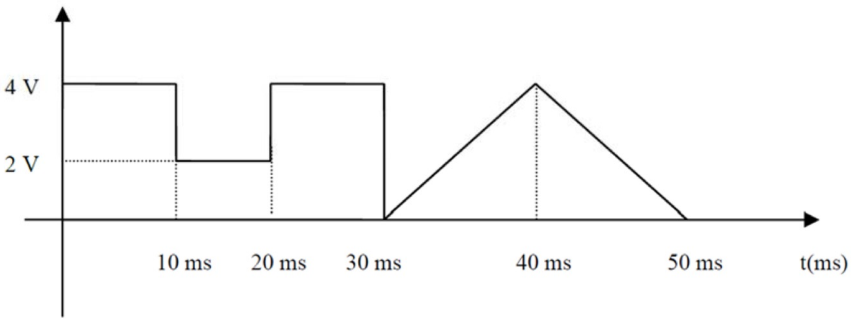
Exercise 1

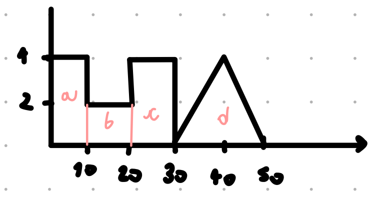
Given the periodic signal s(t) shown in the figure, with period T=50 ms:



Calculate its Fourier series and signal power.

Let us assume that we have a second periodic signal w(t), with the same period as s(t) and equal to s(t) only between 0 and 30 ms, and constant in the rest of the period. Determine what value w(t) should take between 30 and 50 ms to have the component of order 0 in the Fourier series equal to the analogous Fourier series component of s(t).

$T_0 = 50$ ms



CALCULATE FOURIER SERIES AND POWER

$$s(t) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{j2\pi nt}{T_0}}$$

$\big\downarrow$ SPLIT THE SIGNAL IN SIMPLER ONES

$$\left.\begin{aligned} a(t) &= 4\cdot \text{RECT}\left(\tfrac{t-5}{10}\right)\\ b(t) &= 2\cdot \text{RECT}\left(\tfrac{t-15}{10}\right)\\ c(t) &= 4\cdot \text{RECT}\left(\tfrac{t-25}{10}\right)\\ d(t) &= 4\cdot \text{TRIANG}\left(\tfrac{t-40}{20}\right)\end{aligned}\right\} s(t) = a(t)+b(t)+c(t)+d(t)$$

$$c_n^a = 4\cdot 10\cdot \text{SINC}(f10)\,e^{-j2\pi f5} \xrightarrow{f=1/T_0=n/50} c_n^a = \frac{4\cdot 10}{50}\cdot \text{SINC}\left(\frac{n}{50}\cdot 10\right) e^{-j2\pi\frac{n}{50}5} = \frac{4}{5}\text{SINC}\left(\frac{1}{5}n\right)e^{-j\frac{1}{5}\pi n}$$

$$c_n^b = 2\cdot 10\cdot \text{SINC}(f\cdot 10)\,e^{-j2\pi f15} \xrightarrow{f=1/T_0=n/50} c_n^b = \frac{2\cdot 10}{50}\text{SINC}\left(\frac{n}{50}\cdot 10\right) e^{-j2\pi\cdot 15\frac{n}{50}} = \frac{2}{5}\text{SINC}\left(\frac{1}{5}n\right)e^{-j\frac{3}{5}\pi n}$$

$$c_n^b = 4\cdot 10\cdot \text{SINC}(f10)\,e^{-j2\pi f25} \xrightarrow{f=1/T_0=n/50} c_n^c = \frac{4\cdot 10}{50}\text{SINC}\left(\frac{n}{50}\cdot 10\right) e^{-j2\pi 25\cdot\frac{n}{50}} = \frac{4}{5}\text{SINC}\left(\frac{1}{5}n\right)e^{-j\pi n}$$

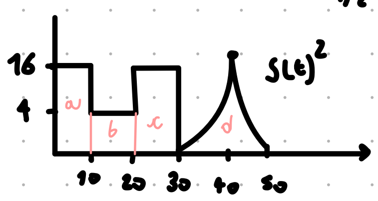
$$x_n^d = 4\cdot \frac{20}{2}\text{SINC}^2\left(f\frac{20}{2}\right) e^{-j2\pi f40} \xrightarrow{f=1/T_0=n/50} c_n^d = \frac{40}{50}\text{SINC}^2\left(\frac{n}{50}\cdot\frac{20}{2}\right) e^{-j2\pi 40\frac{n}{50}} = \frac{4}{5}\text{SINC}^2\left(\frac{1}{5}n\right)e^{-j\frac{8}{5}\pi n}$$

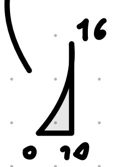
$$c_n = c_n^a + c_n^b + x_n^c + x_n^d = \frac{1}{5}\text{SINC}\left(\frac{1}{5}n\right)\left[4e^{-j\frac{1}{5}\pi n} + 2e^{-j\frac{3}{5}\pi n} + 4e^{-j\pi n}\right] + \frac{4}{5}\text{SINC}^2\left(\frac{1}{5}n\right)e^{-j\frac{8}{5}\pi n}$$

POWER

PARSEVAL THEOREM: TOTAL POWER IS THE SUM OF THE POWER OF EACH COMPONENT

$$P = \sum |c_n|^2 = \frac{1}{T}\int_{-T/2}^{T/2} s^2(t)\,dt = \frac{1}{50}\left[16\cdot 10 + 4\cdot 10 + 16\cdot 10 + 2\left(\frac{160}{3}\right)\right] = \frac{28}{3}$$



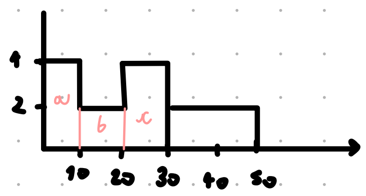


$$y = ax^2 \rightarrow 16 = a100 \rightarrow a = 16/100$$

$$\int_0^{10} at^2\,dt \rightarrow \frac{16}{100}\left[\frac{t^3}{3}\right]_0^{10} = \frac{16}{100}\left(\frac{1000-0}{3}\right) = \frac{160}{3}$$

$$P = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} s(t)^2\,dt = V_0^2 + \sum_{n=1}^{\infty}\frac{V_n^2}{2} = \frac{4^2}{2} + \frac{2^2}{2} + \frac{4^2}{2} + \frac{4^2}{2} = 26$$

w(t)



GIVEN THAT THE COEFFICIENT OF s(t) ARE

$$c_n^{(s)} = \frac{1}{5}\text{SINC}\left(\frac{1}{5}n\right)\left[4e^{-j\frac{1}{5}\pi n} + 2e^{-j\frac{3}{5}\pi n} + 4e^{-j\pi n}\right] + \frac{4}{5}\text{SINC}^2\left(\frac{1}{5}n\right)e^{-j\frac{8}{5}\pi n}$$

THEN

$$c_0^{(s)} = \frac{1}{5}\text{SINC}(0)\left[4\cdot 1 + 2\cdot 1 + 4\cdot 1\right] + \frac{4}{5}\text{SINC}^2(0)\cdot 1$$

$$= \frac{1}{5}[4+2+4] + \frac{4}{5}\cdot 1\cdot 1 = \boxed{\frac{14}{5}}$$

$\longrightarrow$ SINCE THE FIRST PART FROM 0 TO 30 MS IS CONSTANT FOR BOTH s(t) AND w(t) $\longrightarrow$ $c_n^{(w)}$ OF w(t) MUST BE $\frac{4}{5}$ FROM 30 TO 50 ms

$\downarrow$

$w(t) = \frac{4}{5}\ \forall t \in [30;50]$ (A CONSTANT SIGNAL HAS ONLY THE C_0 COMPONENT)

$\text{SINC}(x) = \frac{\text{SIN}(\pi x)}{\pi x}$

AND $\text{SINC}(0) = 1$